2025_Hexaware_Actual_Test_11

Test Summary

- No. of Sections: 3
- No. of Questions: 92
- Total Duration: 130 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. Twenty times a positive integer is less than its square by 96. What is the integer?

- 24
- -4
- -24
- 4

Q2. Find the number of ways a batsman can score a double century only in terms of 4's & 6's?

- 14
- 15
- 16
- 17

Q3. A and B can do a piece of work in 12 days and B and C can do it in 15 days. If all the three work together, it can be finished in 10 days. How long will it take for A and C to do it together?

- 18 days
- 25 days
- 16 days
- 20 days

Q4. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

- 12 days
- 15 days
- 16 days
- 18 days

Q5. Train A sees another train B 60 m ahead of it. After 60 seconds, train A is 90 metres behind. If the speed of train B is 30 km/h, what is the speed of train A?

- 10 km/h
- 12 km/h
- 15 km/h
- 21 km/h

Q6. Deepak takes 24 minutes more to cover a certain distance by decreasing his speed by 25%. What is the time taken by him to cover the distance with his original speed?

- 70 minutes
- 72 minutes
- 75 minutes
- 90 minutes

Q7. A trader aims for a 25% profit on the cost price after allowing a 3.85% discount. By what percentage should he mark his goods above the cost price?

- 125
- 130
- 135
- 140

Q8. Anush gave 40% of his retirement money to his wife. He also gave 20 % of the remaining amount to each of his 3 sons. 50 % of the amount now left was spent on miscellaneous items and the remaining amount of Rs. 1,20,000 was deposited in the bank. How

much money did the Anush get as retirement money?

- 8,00,000
- 12,00,000
- 10,00,000
- 15,00,000

Q9. A dealer offers a cash discount of 20% and still makes a profit of 20%, when he further allows 16 articles to a dozen to a particularly sticky bargainer. How much percent above the cost price were his wares listed?

- 75%
- 80%
- 90%
- 100%

Q10. A sells an article to B at a gain of 25% B sells it to C at a gain of 20% and C sells it to D at a gain of 10%. If D pays Rs. 330 for it, how much did it cost A?

- Rs. 290
- Rs. 275
- Rs. 250
- Rs. 200

Q11. Hari took an educational loan from a nationalized bank for his 2 years course of MBA. He took the loan of Rs.5 lakh such that he would be charged at 7% p.a. at CI during his course and at 9% CI after the completion of the course. He returned half of the amount which he had to be paid on the completion of his studies and remaining after 2 years. What is the total amount returned by Hari?

- Rs. 626255
- Rs. 626277
- Rs. 616266
- Rs. 626288

Q12. A sum of money placed at compound interest doubles itself in 3 years. In how many years will it amount to 8 times itself?

9 years

8 years

27 years

7 years

Q13. There are 49 cards in a box numbered from 1 to 49. Every card is numbered in a distinct manner. What is the probability of picking up a card, which is a multiple of 5 but not a multiple of 10 or 15?

3/49

5/49

1/7

1/49

Q14. There are 49 cards in a box numbered from 1 to 49. Every card is numbered with only 1 number. The probability of picking up a card, the number printed on which is a multiple of 5 but not that of 10 or 15 is _____.

3/49

5/49

1/7

1/49

Q15. 9 persons were invited for a business meeting by Ambani, where the host was seated at a circular table. How many different arrangements are possible if two invites viz., Sahara and Mahindra be seated on either side of the host Ambani?

10080

10800

9200

4600

Q16. A class is composed of two brothers and six other boys. In how many ways can all the boys be seated at a round table so that the two brothers are not seated beside each other?

720

1440

3600

4320

Q17. The arithmetic mean of the scores of a group of students in a test was 64. The brightest 20% of them secured a mean score of 90 and the dullest 20% secured a mean score of 20. The mean score of the remaining 60% is:

72

70

68

65

Q18. Average age of girls in a class is 16 years. If the average of the boys in class is also added, the average becomes 15.5 years. If there were 20 boys in the class with average age 15 years, how many girls were there in the class?

15

40

20

25

Q19. There are two boxes A and B. Box A contains 30 kg of Rice and box B contains 25 kg of wheat. 5 kg of rice was taken from box A and poured into box B, then 12 kg of the mixture was taken from box B and poured into box A. What is the ratio of the quantity of rice in box A to the quantity of wheat in box B?

11:9

7:3

9:5

8:3

Q20. Rs.624.60 is to be divided among A, B and C. If A receives Rs.130 less than C and C receives Rs.25 less than B, then how much will each receive? (in order of A, B and C respectively) (in Rs.)

243.20, 268.20, 113.20

243.20, 113.20, 268.20

268.20, 113.20, 243.20

113.20, 268.20, 243.20

Q21. Suppose that a particular 'TIGER' is written as 'QDFHS'. Find out how 'FISH' will be written in the same code.

GERH

GRHE

GREH

GHRE

Q22. If 'FROZEN' is decoded as 'OFAPSG'. Tick the right option that depicts 'MOLTEN' written in this way.

OFPOMN

OFSMPN

OFUMPN

OFUNPN

Q23. **Directions:**
What will replace the question mark (?) in the following series?

15 8 9 ? 32 82.5 250.5

Q24. **Directions:**
What will replace the question mark (?) in the following series?

15, 19, 83, 119, 631, ?.

Q25. A is the only son of P, P is B's mother's Mother-in-Law. How is A related to B?

- Father
- Brother
- Uncle
- Sister

Q26. A man walks 8m towards the east to reach point P. He turns to his right and walks 5m to reach point Q. Then, he turns to his left and walks 9m to reach point R and he turns to his right and walks 11m to reach point S. Finally he turns to his right and walks 20m to reach point T.

What is the distance between the starting point and Q?

- 13 m
- 9 m
- $\sqrt{89}$ m
- $3\sqrt{27}$ m

Q27. Komal, who is facing North, turns at 60° in the clockwise direction and then 225° in an anticlockwise direction after that she again turns 290° in the clockwise direction. Now towards which direction is she facing?

- North-east
- North-west
- South-east
- South-west

Q28. **The question below consists of a question and two statements numbered (I) and (II) given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and answer.**

How is Sulekha related to Nandini?
I) Sulekha's husband is the only son of Nandini's mother.
II) Sulekha's brother and Nandini's husband are cousins.

- I alone is sufficient while II alone is not sufficient
- II alone is sufficient while I alone is not sufficient
- Either I or II is sufficient

Neither I nor II is sufficient

Both I and II are sufficient

Q29. **The question below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both the statements and give an answer.**

If r and s are integers, is r divisible by 7?
(A) The product rs is divisible by 7.
(B) s is not divisible by 7.

if statement (A) alone is sufficient to solve the question, but statement (B) alone is not.

if statement (B) alone is sufficient to solve the question, but statement (A) alone is not.

if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question.

if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question.

if both the statements taken together are not sufficient and more information is required to solve the question.

Q30. Read the conclusions and then decide which of the given conclusions logically follows from the given statements disregarding the commonly known facts.

Statements:
Some classes are plates
Some plates are trunks
Some trunks are elephants

Conclusions:
I. All elephants being trunks is a possibility
II. All trunks being plates is a possibility

Only I follows

Only II follows

Either I or II follows

Both I and II follows

Q31. **Directions:** In the following question, three conclusions are given and five statements are given as a), b), c), d) and e). From this, you have to take the statements to be true even if they seem to be at variance with commonly known facts and then decide which of the given statement logically follows.

Conclusions:
Some roots are clothes
Some rags are wells is a possibility
Some clothes are rags

a) Statements: All rags are tanks. No rags are clothes. Some tanks are not wells. All wells are roots
b) Statements: No rags are tanks. All tanks are clothes. Some clothes are not well. All wells are roots
c) Statements: All rags are tanks. All tanks are clothes. Some clothes are not well. All wells are roots
d) Statements: Some rags are tanks. All tanks are clothes. Some clothes are well. All wells are roots
e) Statements: Some rags are tanks. All tanks are clothes. No clothes are well. No wells are roots

(a)

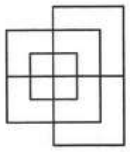
(b)

(c)

(d)

(e)

Q32. Find the minimum number of straight lines required to make the given figure.



13

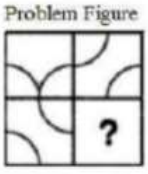
15

17

19

Q33. Select the option that fits in the blank space in the given Problem Figure.

Problem Figure



a 

b 

c 

d 

1

2

3

4

Q34. In the question below, a set of sentences is given which when properly sequenced, form a coherent paragraph. Arrange the sentences in the correct sequence from the answer options.

A. Cotton is a Kharif crop that requires 6 to 8 months to mature.
B. So, in Northern India, it is harvested in December before the winter frost damages crops.
C. Its time of sowing and harvesting differs in different parts of the country.
D. It is so because sowing and harvesting time depend upon climatic conditions.
E. Whereas, in the peninsular part of India, there is no such danger of winter frost.

ACDBE

DCEAB

DEABC

ACEBD

Q35. **In the questions below, each passage consists of six sentences. The first and sixth sentences are given in the beginning. The middle four sentences in each have been removed and jumbled up. These are labeled as P, Q, R, and S. Find out the proper order for the four sentences.**

S1: For some time in his youth Abraham Lincoln was manager for a shop.
P: Then a chance Customer would come.
Q: Young Lincoln's way of keeping shop was entirely unlike anyone else's
R: Lincoln would jump up and attend to his needs and then revert to his reading.
S: He used to lie full length on the counter of the shop eagerly reading a book.
S6: Never before had Lincoln had so much time for reading as had then.

SRQP

QSPR

SQRP

QPSR

Q36. **Select the option that is most nearly OPPOSITE in meaning to the given word:**

BLASÉ (OPPOSITE)

- fiery
- clever
- intriguing
- slim
- ardent

Q37. Choose the appropriate word that answers the description.

A nobleman or person of high position.

- cavalcade
- aristocrat
- bureaucrat
- caricature

Q38. **Select the synonym for the given word.**

VITIATE

- vitaminize
- ventilate
- improve
- spoil

Q39. **Fill up the blank with suitable prepositions and conjunctions from the alternatives given under each sentence.**

No sooner had I reached the office _____ I got the news of your arrival.

- and
- when
- than

since

Q40. **Complete the following sentence by choosing the right option.**

I didn't know you were interested ..... science.

in

for

on

to

Q41. **Complete the following sentence by choosing the right option.**

Suzanne who lives ..... that coffee shop place, came over to visit you this morning.

on

by

at

in

Q42. **Pick out the best alternative for the underlined part of sentence.**

His dog, along with his cat and goldfish, prevent him from taking long trips.

along with his cat and goldfish, prevent

as well as his cat and goldfish, prevents

in addition to his cat and goldfish, are preventing

together with his cat and goldfish, were preventing

Q43. **Read each sentence to find out whether there is any error in it. The error, if any, will be in one part of the sentence. The number of that part is the answer. Mark "No error " as the answer if there is no error in the sentence.**

(1)The children / (2) have been / (3) playing since / (4) two hours

1

2

3

4

No error

Q44. **Fill in the blanks with the right option:**

Sarah and I _____ the car for Mr. Jones last night.

cleaned

cleans

cleaning

have cleaned

Q45. **Fill in the blanks with the suitable option:**

While they _____, they heard a terrific roar.

had been talking

were talking

have been talking

will be talking

Common Content:

Twelve members of a cricket team are sitting in two parallel rows containing 6 people each, in such a way that there is an equal distance between adjacent persons.
In row-1 P, Q, R, S, T, and U are seated and all of them are facing South.
In row-2 A, B, C, D, E and F are seated and all facing North.
Therefore in the given seating arrangement each member seated in a row faces another member of the outer row. S sits third to right of Q, either S or Q sits an extreme end of line. The one who faces Q sits second to right of E. Two people sits between B and F. Neither B nor F sits at an extreme end of the line. The immediate neighbour of B faces the person who sits third to left of P, R and T are immediate neighbours of the each other. C sits second to the left of A. T does not face the immediate neighbour of D.

Q46. Three of the following four are alike in a certain way based on the given arrangement & so form a group. Which is the one that does not belong to that group?

B-T

F-P

C-U

E-Q

Q47. P is related to A in the same way as S is related to B based on the given arrangement. To which of the following is T related to, following the same pattern?

C

D

E

F

Q48. Who amongst the following sits at the extreme ends of the rows?

Q, A

U, F

Q, E

P, D

Q49. How many persons are seated between U and R?

One

Two

Three

Four

Q50. Who amongst the following faces S?

A

B

C

D

Common Content:

Read the following information and answer the questions given below. The six members of a family M , N, O ,P, Q and R are traveling together. N is the son of O but O is not the mother of N. M and O are a married couple. Q is the brother of O. P is the daughter of M. R is the brother of N

Q51. How is Q related to N ?

Maternal uncle

Paternal uncle

Niece

Nephew

Q52. How is P related to Q?

Paternal uncle

Maternal uncle

Niece

Nephew

Common Content:

Read the passage given below and answer the questions that follow:

One of the most dramatic scientific adventures of this century is the exploration of space. We are not content to understand that it is infinite. We want landmarks, so to speak platforms for our satellites, or mathematical slots into which we can aim our space vehicles. We want to comprehend space; define it; in a sense to use it. The other great cosmic reality is time. We may speculate about either end of our earthly existence. We may trust in immortality in the face of incomprehensible death; but, as in our efforts to define space, we must in our definition of time start where we are. All we can know is that man's average ratio of time is three scores and ten years. What we do with our known allotment is what concerns us. Of most immediate concern is what we do with the smaller blocks of time within our grasp: the next week, the next day, the next hour, this very hour.

We all share a common belief that life is too short to be little. Yet the greatest frustration is that so much of life is just that. Perhaps more significant and dramatic than space exploration is an investigation of our use of time. "What folly", said John Howe, "to dread the thought of throwing away life at once, and yet have no regard to throwing it away by parcels and piecemeal."

As with space, we are not content to comprehend time only as infinite. For many people, the pressing questions are "How am I going to manage my time?" Very busy people with many external demands do not have time on their hands. The next hour is very well programmed. This programming or structuring is what people try to achieve, and when they are unable to do it themselves, they look to others to structure time for them. "Tell me what to do." "What shall I do next?" What we need is leadership.

Structure hunger is an outgrowth of recognition hunger, which grew from the initial stroking hunger. The small child does not the necessary comprehension of time to structure it but simply sets about things that feel good, moment to moment. As he gets a little older he learns to postpone gratifications for greater rewards: "I can go outside and make pies with Susie now, but if I wait twenty more minutes and keep my nice dress on, I can go to the shopping center with Daddy." This is basically a problem in structuring time. Which alternative will be more fun? Which will bring a greater reward? As we grow older we have more and more choices. However, the NOT OK position keeps us from exercising these choices as freely as we might think we do.

In our observation of transactions between people, we have been able to establish six types of experiences, which are inclusive of all transactions. They are withdrawal, rituals, activities pastimes, games, and intimacy.

Withdrawal although is not a transaction with another person, can take place, nonetheless, in a social setting. A man having lunch with a group of boring associates more concerned about their own stroking than his many withdraw into the fantasy of the night before when the stroking was good. His body is still at the lunch table, but he isn't. Schoolrooms on a nice spring day are filled with bodies whose occupants are down at the swimming hole, shooting into space on a blazing rocket, or recalling how nice it was kissing under the wisteria. Whenever people withdraw in such fashion it is always certain that the withdrawal keeps them apart from those they are with bodily. This is fairly harmless unless it happens all the time, or unless your wife is talking to you.

A ritual is a socially programmed use of time where everybody agrees to do the same thing. It is safe, that there is no commitment to or involvement with another person, the outcome is predictable, and it can be pleasant in so far as you are in step or doing the right thing. There are worship rituals, greeting rituals, cocktail party rituals, and bedroom rituals. The ritual is designed to get a group of people through

the hour without having to get close to anyone. They may but they don't have to. It is more comfortable to go to a High Church Mass than to attend a revival service where one may be asked, "Are you saved, brother?" Sexual relations are less awkward in the dark for people for whom physical intimacy has no involvement at the level of personality. There is less chance for involvement in throwing a cocktail party than in having a dinner for six. There is little commitment and, therefore little fulfillment. Rituals, like withdrawal, can keep us apart.

An activity, according to Berne, is a common convenient, comfortable, and utilitarian method of structuring time by a project designed to deal with the material of external reality. Common activities are keeping business appointments, doing the dishes, building a house, writing a book, shoveling snow, and studying for exams. These activities, in that they are productive or creative, may be highly satisfying. They may also lead to satisfaction in the future in the activity and there is no need for intimate involvement with another person. There may be, but there does not have to be. Some people use their work to avoid intimacy, working at night at the office instead of coming home, devoting their lives to making a million instead of making friends. Activities, like withdrawal and rituals, can keep us apart.

Q53. "Withdrawal though not a transaction with another person can take place in a social setting": Which of these is true with respect to this statement?

- A man having lunch with a group of boring companions more concerned about their own stroking than his may withdraw into his own fantasy.
- A person withdrawing from his colleagues may be successful in hiding it.
- Whenever people withdraw into their fantasies they are mentally apart, it is harmless if does not happen all the time.
- All of the these

Q54. All of the following are reasons for: "One of the most dramatic scientific adventures of this century is the exploration of space" except

- we are not pleased to understand that it is infinite.
- we want landmarks, say platforms for our satellites or mathematical slots into which we can aim our space vehicles.
- we want to comprehend space.
- we want to define the comprehension of space in a sense to use it.

Q55. "As with space, we are not content to comprehend time only as infinite". Which of the following is true with respect to this statement?

- For many people the important problem is that of time management.
- Many busy people with many external demands do not have time on their hands.
- The programming or structuring of time is difficult for many people, so they look to others, for leadership.
- All of the these.

Q56. The author is most likely a

- professional psychologist
- professional economist
- sociologist
- newspaper editor

Common Content:

Choose the appropriate words to complete the passage

The key _____(1) from the UN's World Population Prospects 2019 report is that national leaders must _____(2) their efforts to raise education, health, and living standards for people everywhere. India is _____(3) to become the most populous country by 2027 surpassing China, and _____(4) 1.64 billion people by 2050; the world as a whole could be home _____(5) 8.5 billion people in just over a decade from now, and the number could go up to 9.7 billion by mid-century. The projections should be viewed in perspective, considering that alarmist Malthusian fears of the inability to provide for more than a billion people on Earth did not come true.

Q57. Choose the appropriate word for the blank numbered (5)

- for
- over
- to
- there

Q58. Choose the appropriate word for the blank numbered (4)

- lead
- host
- hosting
- enable

Q59. Choose the appropriate word for the blank numbered (2)

- redouble
- intensify
- reattempt
- restart

Q60. Choose the appropriate word for the blank numbered (1)

effect

status

report

message

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

```
1  Integer a, b, c
2  Set a=9, b=4, c=15
3  a= (c+9)^b
4  b= (a+9)+a
5  a= 3^b
6  Print a+b+c
```

143

147

156

146

Q2. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=6, q=7, r=6
3  if(p>q || q<r)
4    r = 12+p
5    q=(q+r)+q
6  End if
7  Print p+q+r
```

56

19

55

68

Q3. What will be the output of the following pseudocode?

```
Integer v, a, r
Set v = 5 , a = 64
r = a >> v
r = a - r
Print r
```

64

66

62

68

Q4. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=0,b=11,c=10
for (each c from 4 to 6)
   a=(3+4)+b
   b=a&a
End for
b=(c+c)+b
Print a+b
```

78

69

88

84

Q5. What will be the output of the following pseudocode?

```
Integer p,q,r
Set p=5, q=4, r=9
r=(5+11)+r
q=(p+q)^q
if(9>q || (1-3)>(r+1))
   p=p+r
   r=(p+7)+r
End if
r= (9+7)+q
r=(6&9)^r
Print p+q+r
```

50

59

47

43

Q6. Consider the following pseudocode. Predict the output for the given pseudocode if input n=4.

```
1  function (input n){
2    if (n<2)
3      return n;
4    else  return function(n-1)+function(n-2);
5  }
```

8

7

2

3

Q7. What will be the output of the following pseudocode for a=1 and b=5?

```
1  Integer funn(Integer a, Integer b)
2    if (a&b>0)
3      return 1+funn(b-1,a)
4    Else
5      return a+b
6    End if
7  return b^a
```

19

0

14

6

Q8. What will be the output of the following pseudocode?

```
1  Integer p, q, r
2  Set p= 5, q = 5, r = 10
3  For (each r from 4 to 8)
4    q = (5^7) + p
5    if ((3^4) < r)
6      jump out of the loop
7    Else
```

```
8       p= p+r
9     End if
10  End for
11  Print p+q
```

- 50
- 64
- 57
- 56

Q9. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=4, q=7, r=8
3  q=(9+4)+r
4  r=p+p
5  r=(9^4)+q
6  p=10&q
7  Print p+q+r
```

- 60
- 55
- 52
- 57

Q10. What will be the output of the following pseudocode for a=3 and b=5?

```
1  Integer funn(Integer a, Integer b)
2    if (a>5 || b>5)
3       a=10
4       b=10
5       a=a+10
6       b=b+10
7       a=a+b
8       b=b+a
9       return a+b
10    Else
11      return funn(b+2,a+2)
12    End if
```

- 89
- 101
- 100
- 110

Q11. What will be the output of the following pseudocode?

```
Integer a,b,c,z
Set a=10, b=20, z=1
if((a>>2) > (b>>1))
   z=z+1
end if
if((a>>1) < (b>>2))
   z=z+1
end if
Print z
```

- 1
- 2
- 3
- None of the mentioned options

Q12. What will be the output of the following pseudocode for a=9 and b=3?

```
Integer funn(Integer a, Integer b)
   if (b<a)
      return funn(a-b,b+b)
   End if
   return b-a
```

- -18
- 15
- 0
- 7

Q13. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=8,b=5,c=8
if ((b+c)<(c-b))
   a=(c+c)+a
   b=c+a
   a=(c&b)+c
End if
a=(2+11)+b
Print a+b+c
```

- 32
- 31
- 49

24

Q14. What will be the output of the following pseudocode when **fun(10,12)** is called from the main function?

```
1  Integer function fun(int a, int b)
2  while( a not equals to b)
3    if (a>b)
4     return fun (a-b, b)
5    else
6     return fun (a, b-a)
7    end if
8  end while
9  return a
10 End function fun()
```

3

0

2

4

Q15. What will be the output of the following pseudocode?

```
1  Integer j, m, a[]
2  Set a[]={2,3,5,7,1}
3  m=a[0]
4  For (each j from 0 to 4)
5     if (m<a[j])
6        m=a[j]
7     End if
8  End for
9  Print m
```

7

5

2

10

Q16. What will be the output of the following pseudocode?

```
1  Integer q
2  Set q=0
3  for( each q from 0 to 8)
4    q = q+1
5    if(q EQUALS 6)
6      jump out of the loop
7      Print "Successful"
8    end if
9  end for
10 Print q
```

Successful

Successful
6

6

9

Q17. What will be the output of the following pseudocode for x = 4 and y = 2

```
1  Integer fun (int x, int y)
2    if (x > 1)
3       fun (x - 2, y + 2)
4    end if
5    print y
6  End Function fun()
```

4 5 6

9 7 5

7 6 5

None of the mentioned options

Q18. What will be the output of the following pseudocode for a=4 and b=1?

```
1  Integer funn(Integer a, Integer b)
2    if (a > b && (b == (a > 0)))
3       return b + funn(a - b + a + a + a, b - b - b - b + a)
4    End if
5    if (1 > b && a)
6       return a - b
7    End if
8    return a
9
```

5

16

13

Infinite recursive call

Q19. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=8,b=4,c=15
3  c=b^a
4  a=7+a
5  a=(a&c)+b
6  Print a+b+c
```

- 32
- 28
- 41
- 33

Q20. In a star topology, what is the central point to which all devices are connected?

- Hub
- Router
- Server
- None of the mentioned options

Q21. In a ring topology, how are the devices connected?

- All devices connected to a central hub
- Devices connected in a circular fashion
- Devices connected in a straight line
- Devices connected in a random pattern

Q22. What is the disadvantage of a bus topology?

- Cheap
- Easy to install
- Difficult to troubleshoot
- All of the mentioned options

Q23. Which topology uses a central switch or hub to connect all devices?

- Bus

Ring

Star

Mesh

Q24. What is a characteristic of a bus topology in terms of data transmission?

Full-duplex

Half-duplex

Simplex

Multiplex

Q25. What is the primary purpose of DNS (Domain Name System)?

Dynamic Network Settings

File Transfer

Name Resolution

Email Transmission

Q26. What does HTTP stand for?

Hypertext Transfer Protocol

Hypertext Transmission Protocol

Hypertext Text Protocol

Hypertext Terminal Protocol

Q27. TFTP (Trivial File Transfer Protocol) operates at which layer of the OSI model?

Application Layer

Presentation Layer

Transport Layer

Network Layer

Q28. Which protocol is used for remote login?

SMTP

FTP

TFTP

TELNET

Q29. In TCP, what is the purpose of the sequence number field?

Identifying the source port

Identifying the destination port

Ordering and reordering segments

Specifying the window size

Q30. Which field in the TCP header is used to ensure data integrity?

Acknowledgment Number

Checksum

Window Size

Congestion control mechanism

Section 3 - Coding

Section Summary

- No. of Questions: 2
- Duration: 40 min

Additional Instructions:

None

Q1. **Problem Statement**

You are given an integer array **price,** where **price[i]** is the price of the ith item in a shop.

There is a special discount for items in the shop.

1. If you buy the ith item, then you will receive a discount equivalent to price[j], where j is the minimum index such that j > i and price[j] ≤ price[i].
2. Otherwise, you will not receive any discount at all.

Write a program to calculate the final prices for each item considering this special discount policy using the insertion sort algorithm.

Example 1:

Input:
price = [8, 4, 6, 2, 3]

Output:
4 2 4 2 3

Explanation:
For item 0 with price[0]=8, you will receive a discount equivalent to price[1]=4, therefore, the final price you will pay is 8 - 4 = 4.
For item 1 with price[1]=4, you will receive a discount equivalent to price[3]=2, therefore, the final price you will pay is 4 - 2 = 2.
For item 2 with price[2]=6, you will receive a discount equivalent to price[3]=2, therefore, the final price you will pay is 6 - 2 = 4.
For items 3 and 4, you will not receive any discount at all.

Example 2:

Input:
price = [1, 2, 3, 4, 5]

Output:
1 2 3 4 5

Explanation:
In this case, for all items, you will not receive any discount at all.

Input Format

The first line of input consists of an integer **N,** denoting the number of items in the shop.
The second line consists of **N** space-separated integers, representing the prices of the items in the shop.

Output Format

The output displays **N** space-separated integers, representing the final prices of the items after applying the special discounts.

Refer to the sample output for the formatting specifications.

Constraints

1 ≤ N ≤ 500
1 ≤ prices of the item ≤ 1000

Sample Input

```
5
8 4 6 2 3
```

Sample Output

```
4 2 4 2 3
```

Sample Input

```
5
1 2 3 4 5
```

Sample Output

```
1 2 3 4 5
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2. **Problem Statement**

You are given an array prices where prices[i] is the price of a given stock on the ith day.

You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock.

Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0.

Example 1
Input:
6
7 1 5 3 6 4
Output:
7
Explanation:
Buy on day 2 (price = 1) and sell on day 3 (price = 5), profit = 5-1 = 4.
Then buy on day 4 (price = 3) and sell on day 5 (price = 6), profit = 6-3 = 3.
Total profit is 4 + 3 = 7.

Example 2
Input:
5
7 6 4 3 1
Output:
0
Explanation:
There is no way to make a positive profit, so we never buy the stock to achieve the maximum profit of 0.

Input Format

The first line of input consists of an integer **N,** representing the number of days.
The second line consists of **N** space-separated integers, where each integer represents the stock price on the corresponding day.

Output Format

The output prints the maximum profit that can be achieved.

Refer to the sample output for formatting specifications.

Constraints

In this scenario, the test cases fall under the following constraints:
1 ≤ N ≤ 12
1 ≤ prices[i] ≤ 12

Sample Input

```
6
7 1 5 3 6 4
```

Sample Output

```
7
```

Sample Input

```
5
1 2 3 4 5
```

Sample Output

```
4
```

Sample Input

```
5
7 6 4 3 1
```

Sample Output

```
0
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Answer Key & Solution

Section 1 - Aptitude

Q1

24

Solution

X^2 – 20X = 96

X^2-20X-96 = 0

(X-24)(X+4) = 0

X= 24, -4

positive integer = 24

Q2

17

Solution

Assume a batsman scored X 4's and y 6's

⇒ 4x + 6y = 200

⇒ 2x+3y=100

As x is an integer, y should be a multiple of 2.

If

y = 0, x = 50

y = 2, x = 47

y = 4, x = 44

...

y = 32, x = 2

So total ways are (32-0)/2 + 1 = 17 (if 0 6's are possible) otherwise 16

Q3

20 days

Solution

Work done by A and B in 1 day = 1/12

Work done by B and C in 1 day = 1/15

Work done by A, B and C in 1 day = 1/10

Work done by 2A, 2B, 2C in 1 day = 1/5

Work done by A + 2B + C in 1 day = 1/12 + 1/15 = 3/20

Work done by A and C in 1 day = 1/5 – 3/20 = 1/20

A and C together can do it in 20 days.

Q4

15 days

Solution

A = 20 days - 3 units/day

B = 30 days - 2 units/day

C = 60 days - 1 unit/day

Total work = 60 units

A can do 6 units for 2 days. On third day, B and C assist them to do 6 units

For three days, 12 units is completed.

Time = 60/12 = 5 cycles

Days = 5 x 3 = 15 days.

Q5

21 km/h

Solution

Relative speed = Total distance/Total time = (60 + 90)/60 = 5/2 m/s = 5/2 x 18/5 = 9 km/h

Now, Relative speed = 9 km/h = Speed of train B – Speed of train A

9 = 30 – Speed of train A

Speed of train A = 21 km/h

Q6

72 minutes

Solution

Let the original speed of Deepak be 4x kmph and original time be T hr.

As per question, decreased speed of Deepak = 3x kmph

and now the increased time of Deepak = (T + (24/60))

= T + 0.4 hours.

So, 4x * T = 3x * (T + 0.4)

T = 1.2hour = 72 minutes

Q7

130

Solution

Let the Cost Price (CP) be 100:

CP=100

Calculate the Selling Price (SP) for a 25% profit on the Cost Price:

SP=CP+25% of CP=100+25=125

Given a discount of 3.85%, we know that the Selling Price is 96.15% of the Marked Price (MP):

96.15% of MP=SP=125

0.9615×MP=125

Calculate the Marked Price (MP):

MP= 1250/0.9615=130

Q8

10,00,000

Solution

Let the retirement money be Rs. 100 n

=> Money given to wife = 40 % of 100 n = 40 n,

Balance = 60 n

=> Money given to 3 sons = 3 x (20 % of 60 n) = 3 x 12 n = 36 n, Balance = 24 n

=> Money spent on miscellaneous items = 50 % of 24 n = 12 n, Balance = 12 n

Now, this remaining 12 n is the money deposited in the bank, i.e., Rs. 1,20,000

=> 12 n = 1,20,000

=> n = 10,000

Therefore, Anush's retirement money = 100 n = Rs. 10,00,000

Q9

100%

Solution

Let the CP of the article be Rs. x, since he earns a profit of 20%, hence SP = X + 20% of X = 1.2x

It is given that he incurs loss by selling 16 articles at the cost of 12 articles [loss = 16−1216

= 25%]

His selling price = SP - 25% of SP = SP × 0.75

Hence, SP × 0.75 = 1.2X

This SP is arrived after giving a discount of 20% on MP.

Let MP = Y

Y - 20% of Y = SP

0.80Y = 1.6X

Y = 2X

It means that the article has been marked 100% above the cost price. Or Marked Price was twice of cost price.

Q10

Rs. 200

Solution

Let Cost Price for A was 100

Then CP for B = 100 + 25% of 100 = 125

CP for C = 125 + 20% of 125 = 150

CP for D = 150 + 10% of 150 = 165

But, D pay Rs. 330, Then it must be equal to

165 = 330

1 = 330/165

100 = (330×100)/165=200

Thus, CP for A = Rs. 200

Q11

Rs. 626288

Solution

5,00,000 x $(1.07)^2$ = 572450

Returned amount = 286225

After two years = 286225 x $(1.09)^2$ = 340063

Total amount = 286225 + 340063 = 626288

Q12

9 years

Solution

Let the principal be Rs. 100

In 3 years it will be Rs. 200

In 6 years it will be Rs. 400

In 9 years it will be Rs. 800

In 9 years the amount to 8 times itself.

Q13

3/49

Solution

According to the given information,

Card numbers which are multiples of 5 are 5, 10, 15, 20, 25, 30, 35, 40, 45

Card numbers which are multiples of 10 or 15 are 10, 15, 20, 30, 40, 45

The latter ones are to be ruled out from the former ones.

⇒ So, the only card numbers we are left with are 5, 25, 35.

∴ Probability of picking up a card, the number on which is a multiple of 5 but not that of 10 and 15 = 3/49

ans option 1

Q14

3/49

Solution

According to the given information,

Card numbers which are multiples of 5 are 5, 10, 15, 20, 25, 30, 35, 40, 45

Card numbers which are multiples of 10 or 15 are 10, 15, 20, 30, 40, 45

The latter ones are to be ruled out from the former ones.

⇒ So, the only card numbers we are left with are 5, 25, and 35.

∴ Probability of picking up a card, the number on which is a multiple of 5 but not that of 10 and 15 = 3/49

Q15

10080

Solution

Assuming Sahara, Ambani and Mahindra as a single personality, then there are total (10 – 3) + 1 = 8 person,

which can be arranged in 7! ways.

Also Sahara and Mahindra can be mutually permuted in 2 ways.

Therefore required number of arrangements = 2 × 7! = 10080.

Q16

3600

Solution

Total number of arrangements = (N-1)! ie 7!

If 2 brothers are together, then the number of ways = 6! × 2!

Required number of ways Two brothers are not together = 7! – 6! 2! = 5040 – 1440 = 3600.

Take 1 person from 6 and fix him and 5 others can be arranged in -- 5! ways=120

There are 6 places left in which 2 brothers can sit.

So they can choose any 2 places from 6 - 6C2 ways=15

2 brothers can arrange themselves in 2! ways=15*2=30

Total ways=120*30=3600

Q17

70

Solution

Let the total number of students be 100

Given,

⇒ Average of 100 students = 64

⇒ Total score of 100 students = 64 × 100 = 6400

Then,

⇒ Total score of first 20 students = 20 × 90 = 1800

⇒ Total score of last 20 students = 20 × 20 = 400

⇒ Total score of remaining 60 students = 6400 – (1800 + 400) = 4200

⇒ Average of 65 students = 4200/60 = 70

Q18

20

Solution

Let x girls in the class, so

16x + 20*15 = 15.5 (x+20) => x =20

Q19

9:5

Solution

BOX A BOX B

rice wheat

30 25

(-5) (+5)

25 30

wheat rice

25 : 5

5 : 1

Total 12 kg taken

10 : 2

Hence, Rice in box A: Wheat in box B

27 : (25-10)

27 :15 = 9 : 5

Q20 113.20, 268.20, 243.20

Solution

A + B + C = 624.6 ....(1)

A = C – 130 ....(2)

C = B – 25 ....(3)

=> (C – 130) + (C + 25) + C = 624.60.

=> 3C = 624.60 + 105.

=> C = Rs. 243.20, B = Rs. 268.20, A = Rs.113.20.

Q21 GRHE

Solution

To solve this coding-decoding question, you have to reverse the word and move each letter -1. The reverse order of FISH is HSIF, now subtract 1 from each alphabet of HSIF. The code for FISH is GRHE.

Q22 OFUMPN

Solution

No Solution

Q23 15

Solution

Series Pattern	Given Series
15	15
$15 \times \frac{1}{2} + \frac{1}{2} = 8$	8
$8 \times 1 + 1 = 9$	9
$9 \times \frac{3}{2} + \frac{3}{2} = 15$	15
$15 \times 2 + 2 = 32$	32
$32 \times \frac{5}{2} + \frac{5}{2} = 82.5$	82.5
$82.5 \times 3 + 3 = 250.5$	250.5

Q24 731

Solution

Q25 Father

Solution

A is son of P.

P is B's mother's mother-in-law which means B is the child of A.

So A is the father of B.

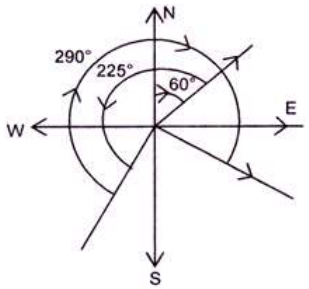
Q26 √89 m

Solution

No Solution

Q27 South-east

Solution



The path traversed by Komal is as follows.

The total angle rotated by Komal is (290° + 60° − 225°) = 125°

∴ 125° clockwise from North will be south-east

∴ Komal is facing towards South-east.

Q28 Either I or II is sufficient

Solution

From I, we conclude that Sulekha is the wife of Nandini's mother's only son i.e. Nandini's brother. Thus, Sulekha is Nandini's sister-in-law.

From II, we conclude that Sulekha is the cousin of Nandini's husband, which implies that Sulekha is Nandini's sister-in-law.

Q29 if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question.

Solution

Statement (A): If a product rs is divisible by the prime number 7, then either r is divisible by 7 or s is divisible by 7.

Hence, (A) alone implies that either r or s is divisible by 7, but it is not sufficient to determine that r is divisible by 7.

Statement (B) alone is clearly not sufficient, since no information is given about r.

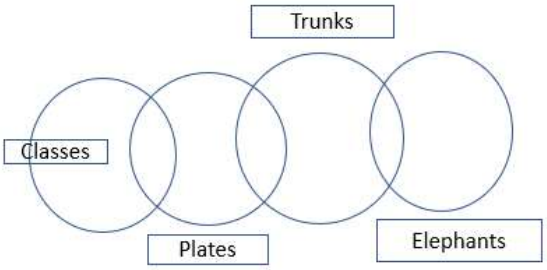
After combining if rs is divisible by 7 and s is not divisible by 7, then r is divisible by 7.

The two statements together are sufficient.

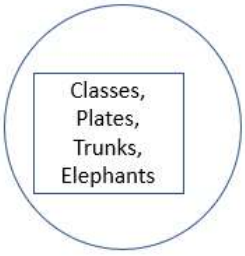
Q30 Both I and II follows

Solution

Case 1 :



Case 2 :



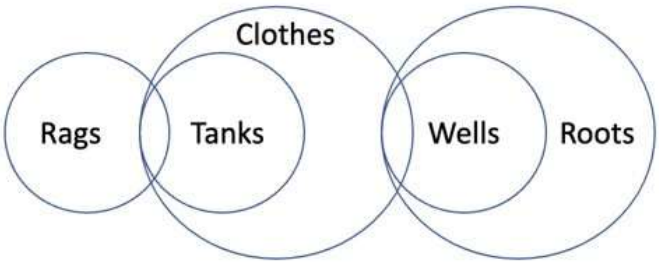
Both I and II are a possibility in case 2.

Hence, I and II follows

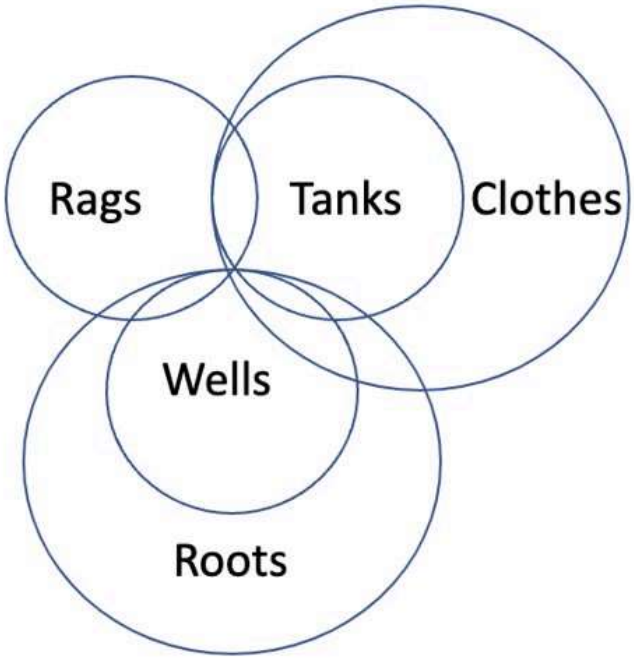
Q31 (d)

Solution

Case I:



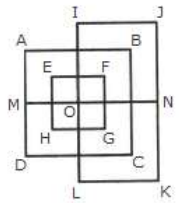
Case II:



Q32 13

Solution

The figure may be labelled as shown.



The horizontal lines are IJ, AB, EF, MN, HG, DC and LK i.e. 7 in number.

The vertical lines are AD, EH, IL, FG, BC and JK i.e. 6 in number.

Thus, there are 7 + 6 = 13 straight lines in the figure.

Q33

1

Solution

By looking at the options the option that fills the empty space at the bottom and has connectivity with the other parts is the answer.



Q34

ACDBE

Solution

The correct sequence is ACDBE.

(A) starts the paragraph and mentions cotton - a Kharif crop, and tells how long the crop takes to mature.

(B) does not follow (A) as 'so' in the sentence implies that (B) is an elaboration of some explanation given previously. 'Elaboration' means the addition of more detail concerning what has already been said.

(C) follows (A) as it informs that sowing and harvesting of cotton are done at different times in different parts of the country. (D) follows and gives the reason behind this (i.e., dependence upon climatic conditions).

(B) and (E) follow and inform about the harvesting pattern in Northern and peninsular parts of India. (B) follows (D) since it starts with 'so' (which is the same as 'therefore') and informs that the crop is harvested in northern parts in December before the winter frost.

'Whereas' in (E) implies it will mention something opposite to (B), which it does by informing that there is no issue of winter frost in peninsular India. Hence, (E) follows (B).

Hence, the course of the passage: Cotton - A Kharif crop and the time it takes to mature (A), Different sowing and harvesting times in different parts of the country (C), reason behind it (D), the scenario in northern parts of India (8), and the opposite scenario in peninsular India (E)

Hence, the sequence is ACDBE

Q35 QSPR

Solution

The sequence QSPR fits the context by first introducing the unique manner in which young Lincoln managed the shop (Q), followed by the description of his way of tending to a chance customer after reading (S). Then, a chance customer's arrival is mentioned (P), leading to Lincoln attending to their needs and returning to his reading (R). This sequence creates a logical flow by introducing Lincoln's atypical approach to managing the shop.

The correct answer is Option (2)

Q36 ardent

Solution

Blase means unimpressed with or indifferent to something . Fiery means passionate.Intriguing means arousing one's curiosity. Ardent means very enthusiastic or passionate. So ardent is the answer.

Q37 aristocrat

Solution

An aristocrat is a member of the highest class in certain societies, typically comprising people of noble birth holding hereditary titles and offices.

Q38 spoil

Solution

Vitiate means to impair the quality of; make faulty; spoil.

Q39 than

Solution

The correct phrase is "No sooner had I reached the office **than I** got the news of your arrival." The expression "no sooner... than" is used to indicate that one event happened immediately after another.

Q40

in

Solution

No Solution

Q41

by

Solution

No Solution

Q42

as well as his cat and goldfish, prevents

Solution

The subject in the sentence is his dog (singular), along with (prepositional phrase) only provides additional information , hence the verb should be singular. The **verb** in the given sentence should also be **singular** (prevents) and hence the answer is **as well as cat and goldfish,prevents**

Q43

3

Solution

Error is in part 3. 'Since' is used with a specific month, year or a period in the past. Use 'for' with a number of hours, days, months, years. In this context, a numbers of hours is mentioned. Hence 'since' is replaced by 'for'.

Q44

cleaned

Solution

Keyword - last night

Tense to be used - Simple past tense

Answer - cleaned

Q45

were talking

Solution

The tense used is past continuous

The subject is plural

The right answer is ***were talking***.

Q46. E-Q

Solution

P	U	S	T	R	Q
C	F	A	E	B	D

Except EQ , all the other pairs are placed immediate diagonal to each other.

Q47. D

Solution

P	U	S	T	R	Q
C	F	A	E	B	D

T is related to D.

Q48. P, D

Solution

P	U	S	T	R	Q
C	F	A	E	B	D

P and D sits at the extreme end of the rows.

Q49. Two

Solution

P	U	S	T	R	Q
C	F	A	E	B	D

Two persons sits between U and R.

Q50. A

Solution

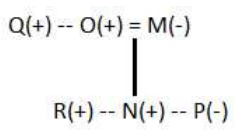
P	U	S	T	R	Q
C	F	A	E	B	D

A faces S.

Q51. Paternal uncle

Solution

Male(+), Female(-), Couple(=), Siblings(--)

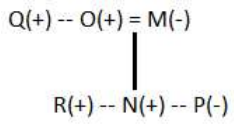


is paternal uncle of N.

Q52. Niece

Solution

Male(+), Female(-), Couple(=), Siblings(--)



P is niece of Q.

Q53. All of the these

Solution

All the three options have been clearly discussed in para 6.

Q54. we want to define the comprehension of space in a sense to use it.

Solution

Options (we are not pleased to understand that it is infinite.), (we want landmarks, say platforms for our satellites or mathematical slots into which we can aim our space vehicles.) & (we want to comprehend space.) are clearly given in para 1.

Hence, by exclusion, the answer is (we want to define the comprehension of space in a sense to use it.).

Q55. All of the these.

Solution

All the three options are indicated in para 3.

Q56. sociologist

Solution

Option (professional economist) is clearly absurd.

It is not a news item, therefore reject option (newspaper editor).

Option (professional psychologist) comes close but can be discarded as no disease process or mental illness is discussed.

The passage talks of types of social activities, hence the answer is (sociologist).

Q57. to

Solution

No Solution

Q58. host

Solution

"host" is the correct option from the given options.

Q59. redouble

Solution

"redouble" is the most suitable from the given options.

Q60. message

Solution

"message" is the most appropriate word from the given options.

Section 2 - Pseudocode and Computer Fundamentals

Q1 146

Solution

Initially a=9, b=4, c=15, then sequence of operation gets executed and update a=66,b=65. So 66+65+15=> 146.

Q2 19

Solution

Initially p=6, q=7, r=6, if condition gets failed so 6+7+6=>19 gets printed.

Q3 62

Solution

Initially v = 5 ,a = 64 , r = 64>>5 => 2

Then r =64-2 => 62.

Q4 78

Solution

For c = 4, a = (3 + 4) + 11 = 18, b = a & a = 18 & 18 = 18.

For c = 5, a = (3 + 4) + 18 = 25, b = a & a = 25 & 25 = 25.

For c = 6, a = (3 + 4) + 25 = 32, b = a & a = 32 & 32 = 32.

After the loop, c is set to 7, and then b = (7 + 7) + 32 = 46, and a = 32.

Thus, the output of a + b will be 32 + 46 = 78.

Q5

47

Solution

Initially p=5,q=4,r=9

r=(5+11)+9 => 25

q=(5+4)^4 => 13

If block condition gets failed then

r= (9+7)+13 => 29

r=(6&9)^29 =>29

Finally 5+13+29 => 47.

Q6

3

Solution

Pseudocode is used to find the Fibonacci of the number. Fibonacci(4) = 3.

Q7

6

Solution

In funn(1,5) => return 1+funn(4,1)

In funn(4,1) => return 5

In funn(1,5) => return 1+5 => 6 returned to main class and gets printed.

Q8

56

Solution

Initially p= 5, q = 5, r = 10

When r=4,q= 7 then else block executes and update p= 9

When r=5,q= 11 then else block executes and update p= 14

When r=6,q= 16 then else block executes and update p= 20

When r=7,q= 22 then else block executes and update p= 27

When r=8 if block condition gets true and break the loop. So 27+29=56.

Q9

55

Solution

Initially p=4,q=7,r=8 after performing sequence of operation p=0,q=21,r=34 so 0+21+34=>55

Q10

100

Solution

In funn(3, 5) , else block executes and call funn(7,5). In funn(7, 5) , if block executes and update a=40,b=60. So 40+60=100.

Q11

1

Solution

n right shift by 1 will always gives answer as n/2. Here both if block gets failed so 1 gets printed.

Q12

0

Solution

In funn(9,3)= 3<9 (True) return funn(6,6).

In funn(6,6) = return 0 to funn(9,3). So gets printed.

Q13

31

Solution

If block executed and update a=18,b=5 and c=8. So 18+5+8 = 31.

Q14

2

Solution

No Solution

Q15

7

Solution

If block executes when j=1,2,3 and finally update m=7.

Q16

Successful

6

Solution

When q is 6 ctrl jump out of the loop and print 6.

Q17

None of the mentioned options

Solution

When fun(4,2) => 4>1 so if block executes and call fun(2,4).

When fun(2,4) => 2>1 so if block executes and call fun(0,6).

When fun(0,6) => 0>1 (false) so print statement executes and print 6. Then function call backtrack to fun(2,4) and executes print statement and print 4.

Then again function call backtrack to fun(4,2) and executes print statement and print 6. So output is 6 4 2.

Q18

16

Solution

No Solution

Q19

32

Solution

c=4^8 = 12 , a=7+8 = 15, a= (15&12)+4 =16, 16+4+12=32.

Q20

Hub

Solution

No Solution

Q21

Devices connected in a circular fashion

Solution

No Solution

Q22

Difficult to troubleshoot

Solution

No Solution

Q23

Star

Solution

No Solution

Q24 Half-duplex

Solution

No Solution

Q25 Name Resolution

Solution

No Solution

Q26 Hypertext Transfer Protocol

Solution

No Solution

Q27 Application Layer

Solution

No Solution

Q28 TELNET

Solution

No Solution

Q29 Ordering and reordering segments

Solution

No Solution

Q30 Checksum

Solution

No Solution

Section 3 - Coding

Q1

Test Case

Input	Output
6 15 10 12 18 14 20	5 10 12 4 14 20

Weightage - 10

Input

```
7
25 30 20 35 28 23 40
```

Output

```
5 10 20 7 5 23 40
```

Weightage - 15

Input

```
8
40 60 20 50 70 30 80 90
```

Output

```
20 40 20 20 40 30 80 90
```

Weightage - 15

Input

```
10
100 90 110 120 80 130 140 70 150 160
```

Output

```
10 10 30 40 10 60 70 70 150 160
```

Weightage - 25

Input

```
3
5 3 7
```

Output

```
2 3 7
```

Weightage - 10

Input

```
9
200 150 300 180 250 120 400 220 100
```

Output

```
50 30 120 60 130 20 180 120 100
```

Weightage - 25

Sample Input

```
5
8 4 6 2 3
```

Sample Output

```
4 2 4 2 3
```

Sample Input

```
5
1 2 3 4 5
```

Sample Output

```
1 2 3 4 5
```

Solution

```cpp
#include <iostream>

void insertionSort(int prices[], int n) {
    for (int i = 0; i < n; ++i) {
        int min_index = -1;
        for (int j = i + 1; j < n; ++j) {
            if (prices[j] <= prices[i]) {
                min_index = j;
                break;
            }
        }

        if (min_index != -1) {
            prices[i] -= prices[min_index];
        }
    }
}

int main() {
    int n;
    std::cin >> n;

    int prices[n];
    for (int i = 0; i < n; ++i) {
        std::cin >> prices[i];
    }

    insertionSort(prices, n);

    for (int i = 0; i < n; ++i) {
        std::cout << prices[i] << " ";
    }

    return 0;
}
```

```c
#include <stdio.h>

void insertionSort(int prices[], int n) {
    for (int i = 0; i < n; ++i) {
        int min_index = -1;
        for (int j = i + 1; j < n; ++j) {
            if (prices[j] <= prices[i]) {
                min_index = j;
                break;
            }
        }

        if (min_index != -1) {
            prices[i] -= prices[min_index];
        }
    }
}

int main() {
    int n;
    scanf("%d", &n);

    int prices[n];
    for (int i = 0; i < n; ++i) {
        scanf("%d", &prices[i]);
    }

    insertionSort(prices, n);

    for (int i = 0; i < n; ++i) {
        printf("%d ", prices[i]);
    }

    return 0;
}
```

Q2

Test Case

Input

```
6
7 6 4 3 1 2
```

Output

```
1
```

Weightage - 10

Input

```
7
3 1 2 4 1 5 6
```

Output

```
8
```

Weightage - 10

Input

```
8
5 6 7 8 7 6 5 4
```

Output

```
3
```

Weightage - 15

Input

```
9
7 4 5 8 3 6 2 1 9
```

Output

```
15
```

Weightage - 15

Input

```
10
10 3 5 7 4 8 3 2 1 9
```

Output

```
16
```

Weightage - 25

Input

```
12
5 3 8 2 10 7 9 4 1 6 12 11
```

Output

```
26
```

Weightage - 25

Sample Input

```
6
7 1 5 3 6 4
```

Sample Output

```
7
```

Sample Input

```
5
1 2 3 4 5
```

Sample Output

```
4
```

Sample Input

```
5
7 6 4 3 1
```

Sample Output

```
0
```

Solution

```
#include <iostream>
#include <algorithm>

int maxProfit(int prices[], int n) {
    int max_profit = 0;  // Initialize the maximum profit to zero

    for (int i = 1; i < n; ++i) {
        // If the current price is higher than the previous day's price, add to the profit
        if (prices[i] > prices[i - 1]) {
            max_profit += prices[i] - prices[i - 1];
        }
    }
```

```
    return max_profit;
}

int main() {
    int n;
    std::cin >> n;

    int prices[n];
    for (int i = 0; i < n; ++i) {
        std::cin >> prices[i];
    }

    int result = maxProfit(prices, n);
    std::cout << result << std::endl;

    return 0;
}
```

```
import java.util.Scanner;

class MaxProfit {
    public static int maxProfit(int[] prices, int n) {
        int maxProfit = 0;  // Initialize the maximum profit to zero

        for (int i = 1; i < n; ++i) {
            // If the current price is higher than the previous day's price, add to the profit
            if (prices[i] > prices[i - 1]) {
                maxProfit += prices[i] - prices[i - 1];
            }
        }

        return maxProfit;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();

        int[] prices = new int[n];
        for (int i = 0; i < n; ++i) {
            prices[i] = scanner.nextInt();
        }

        int result = maxProfit(prices, n);
        System.out.println(result);

        scanner.close();
    }
}
```

```
#include <stdio.h>

int maxProfit(int prices[], int n) {
    int max_profit = 0;  // Initialize the maximum profit to zero

    for (int i = 1; i < n; ++i) {
        // If the current price is higher than the previous day's price, add to the profit
        if (prices[i] > prices[i - 1]) {
            max_profit += prices[i] - prices[i - 1];
        }
    }
```

```
    return max_profit;
}

int main() {
    int n;
    scanf("%d", &n);

    int prices[n];
    for (int i = 0; i < n; ++i) {
        scanf("%d", &prices[i]);
    }

    int result = maxProfit(prices, n);
    printf("%d\n", result);

    return 0;
}
```